Calorimetry lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use an energy balance to analyze mixing and heat transfer.

SUCCESS CRITERIA - I can...

1. Identify hot and cold samples and their masses.
2. Balance energy gained and lost in an ideal model.
3. Discuss heat exchange with the container and surroundings.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg sample has c=500 J/(kg·°C) and warms by 7°C without changing phase. Find Q.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q_{hot} + Q_{cold} = 0$; $Q = mc(T_f - T_i)$.
Ideal isolated mixing, no phase change. Heat lost is negative; heat gained is positive. Real surroundings may exchange energy.

Worked reasoning: Two equal masses of the same liquid at 20°C and 40°C mix in an insulated container. The final 30°C makes the hot sample lose the same energy the cool sample gains.

Your turn: Draw heat-transfer arrows and show the two equal temperature changes.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Equal masses of the same liquid at 20°C and 60°C are mixed in an ideal insulated container. Predict final temperature and justify.

20°C

60°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

In a mixing trial, the warm water loses 300 J. First assume only the cold water receives that energy. Then consider a cup that absorbs 50 J. Draw both accounts and find the cold water's energy gain in each case.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

05 Explain your reasoning

Plan a table for masses, initial temperatures, final temperature and a heat-loss observation. Why should the final temperature be measured promptly after mixing?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A spring stores 4 J at a given compression. Within its Hooke’s-law range, find the energy at 6 times that compression.
Mark equilibrium, displacement and restoring-force direction.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

07 Exit check

An ideal mixture of 30°C and 70°C samples reaches 80°C with no energy added. Is that plausible? Explain 0.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Mix one cup of 20°C water with two equal-size cups of 50°C water, neglecting losses. Predict whether final temperature is above or below 35°C, then calculate it.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3